

ECEN 743: Reinforcement Learning

Policy Gradient Theorem

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References

- [SB, Chapter 13]
- [MMT, Chapter 12]
- [AJKS, Section 3]

Reinforcement Learning Questions

- How do we learn the value of a policy π ?
- How do we learn the optimal action-value function Q^* ?
- How do we learn the optimal policy π^* ?

... without the knowledge of the model P

- Need to **learn** from the observed sequence of states, actions, and rewards

Finding the Optimal Policy Parameter

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- Can we develop (stochastic) gradient algorithms for solving the above problem?

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- Tabular representation

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Consider the basis function $\phi(s, a) \in \mathbb{R}^d$, $d \ll |\mathcal{S}||\mathcal{A}|$. Let $\theta \in \mathbb{R}^d$. Then,

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- Gaussian policies

$$\pi(s, a) = \mathcal{N}(\mu_\theta(s), \Sigma_\theta(s))$$

Gradient of the Value Function

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$$\theta \leftarrow \theta + \alpha \widehat{\nabla_{\theta} V_{\pi_{\theta}}}$$

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Policy Gradient Theorem!

Probability of a Trajectory

- Let τ be a trajectory generated using π_θ ,

$$\tau = (s_0, a_0, s_1, a_1, \dots, s_T, a_T), \quad a_t \sim \pi_\theta(s_t, \cdot), s_{t+1} \sim P(\cdot | s_t, a_t), s_0 \sim \mu_0(\cdot)$$

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- We can extend this to arbitrary long trajectories, $\tau = (s_t, a_t)_{t \geq 0}$,

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Gradient of the Value Function: Form 1

- Let $\tau = (s_t, a_t)_{t \geq 0}$, $s_0 \sim \mu_0(\cdot)$. Denote the cumulative discounted reward of trajectory τ as

$$R(\tau) = \sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)$$

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Theorem (Policy Gradient Theorem - Form 1)

$$\nabla V_{\pi_{\theta}} = \mathbb{E}_{\tau \sim P_{\text{traj}, \theta}(\cdot)} \left[R(\tau) \sum_{t=0}^{\infty} \nabla \log \pi_{\theta}(s_t, a_t) \right]$$

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- Gradient above does not depend on the gradient of the (stationary) distribution induced by the policy!

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$$\nabla V_{\pi_{\theta}} = \nabla \mathbb{E}_{\tau \sim P_{\text{traj}, \theta}(\cdot)} [R(\tau)] = \nabla \sum_{\tau} R(\tau) P_{\text{traj}, \theta}(\tau)$$



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$$\begin{aligned}\nabla V_{\pi_{\theta}} &= \nabla \mathbb{E}_{\tau \sim P_{\text{traj}, \theta}(\cdot)} [R(\tau)] = \nabla \sum_{\tau} R(\tau) P_{\text{traj}, \theta}(\tau) \\&= \sum_{\tau} R(\tau) \nabla P_{\text{traj}, \theta}(\tau) = \sum_{\tau} R(\tau) P_{\text{traj}, \theta}(\tau) \nabla \log P_{\text{traj}, \theta}(\tau) \\&= \sum_{\tau} R(\tau) P_{\text{traj}, \theta}(\tau) \nabla \log \left(\mu_0(s_0) \prod_{t=0}^{\infty} \pi_{\theta}(s_t, a_t) P(s_{t+1} | s_t, a_t) \right) \\&= \sum_{\tau} R(\tau) P_{\text{traj}, \theta}(\tau) \nabla \left(\sum_{t=0}^{\infty} \log \pi_{\theta}(s_t, a_t) \right)\end{aligned}$$



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 - ▶ Variance reduction for PG is an active area of research

Visitation Distribution

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- It is straight forward to show that

$$\rho_{\pi}(x, a) = \rho_{\pi}(x) \pi(x, a)$$

Visitation Distribution and Value Function

Lemma

$$V_{\pi} = \frac{1}{(1-\gamma)} \mathbb{E}_{(s,a) \sim \rho_{\pi}(\cdot, \cdot)} [r(s, a)] = \frac{1}{(1-\gamma)} \sum_{(s,a)} \rho_{\pi}(s, a) r(s, a)$$

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□

Gradient of the Value Function: Form 2

Theorem (Policy Gradient Theorem - Form 2)

$$\nabla V_{\pi_{\theta}} = \frac{1}{(1 - \gamma)} \mathbb{E}_{(s,a) \sim \rho_{\pi_{\theta}}(\cdot, \cdot)} [Q_{\pi_{\theta}}(s, a) \nabla \log \pi_{\theta}(s, a)]$$

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Proof of Policy Gradient Theorem: Form 2

Proof:

$$\begin{aligned}\nabla V_{\pi_{\theta}} &= \sum_s \mathbb{P}(s_0 = s) \nabla V_{\pi_{\theta}}(s) \\&= \nabla \mathbb{E}_{s_0 \sim \mu_0} [V_{\pi_{\theta}}(s_0)] = \mathbb{E}_{s_0 \sim \mu_0} [\nabla \sum_{a_0} \pi_{\theta}(s_0, a_0) Q_{\pi_{\theta}}(s_0, a_0)] \\&= \mathbb{E}_{s_0 \sim \mu_0} [\sum_{a_0} \nabla \pi_{\theta}(s_0, a_0) Q_{\pi_{\theta}}(s_0, a_0) + \sum_{a_0} \pi_{\theta}(s_0, a_0) \nabla Q_{\pi_{\theta}}(s_0, a_0)] \\&= \mathbb{E}_{s_0 \sim \mu_0} [\sum_{a_0} \pi_{\theta}(s_0, a_0) \nabla \log \pi_{\theta}(s_0, a_0) Q_{\pi_{\theta}}(s_0, a_0)] + \mathbb{E}_{s_0 \sim \mu_0} [\sum_{a_0} \pi_{\theta}(s_0, a_0) \nabla Q_{\pi_{\theta}}(s_0, a_0)] \\&= \mathbb{E}_{s_0 \sim \mu_0} \mathbb{E}_{a_0 \sim \pi_{\theta}(s_0, \cdot)} [\nabla \log \pi_{\theta}(s_0, a_0) Q_{\pi_{\theta}}(s_0, a_0)] + \mathbb{E}_{s_0 \sim \mu_0} \mathbb{E}_{a_0 \sim \pi_{\theta}(s_0, \cdot)} [\nabla Q_{\pi_{\theta}}(s_0, a_0)] \\&= \mathbb{E}_{s_0 \sim \mu_0} \mathbb{E}_{a_0 \sim \pi_{\theta}(s_0, \cdot)} [\nabla \log \pi_{\theta}(s_0, a_0) Q_{\pi_{\theta}}(s_0, a_0)] + \\&\quad \mathbb{E}_{s_0 \sim \mu_0} \mathbb{E}_{a_0 \sim \pi_{\theta}(s_0, \cdot)} [\nabla (r(s_0, a_0) + \gamma \mathbb{E}_{s_1 \sim P(\cdot | s_0, a_0)} [V_{\pi_{\theta}}(s_1)])] \\&= \mathbb{E}_{s_0 \sim \mu_0} \mathbb{E}_{a_0 \sim \pi_{\theta}(s_0, \cdot)} [\nabla \log \pi_{\theta}(s_0, a_0) Q_{\pi_{\theta}}(s_0, a_0)] + \\&\quad \gamma \mathbb{E}_{s_0 \sim \mu_0} \mathbb{E}_{a_0 \sim \pi_{\theta}(s_0, \cdot)} \mathbb{E}_{s_1 \sim P(\cdot | s_0, a_0)} [\nabla V_{\pi_{\theta}}(s_1)] \\&= \sum_{(s,a)} \mathbb{P}(s_0 = s, a_0 = a) [\nabla \log \pi_{\theta}(s, a) Q_{\pi_{\theta}}(s, a)] + \gamma \sum_s \mathbb{P}(s_1 = s) \nabla V_{\pi_{\theta}}(s)\end{aligned}$$

Proof of Policy Gradient Theorem: Form 2

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Proof:

$$\begin{aligned} &= \sum_{(s,a)} \mathbb{P}(s_0 = s, a_0 = a) [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] \\ &+ \gamma \sum_{(s,a)} \mathbb{P}(s_1 = s, a_1 = a) [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] + \gamma^2 \sum_s \mathbb{P}(s_2 = s) \nabla V_{\pi_\theta}(s) \end{aligned}$$



Proof of Policy Gradient Theorem: Form 2

Proof:

$$\begin{aligned} &= \sum_{(s,a)} \mathbb{P}(s_0 = s, a_0 = a) [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] \\ &+ \gamma \sum_{(s,a)} \mathbb{P}(s_1 = s, a_1 = a) [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] + \gamma^2 \sum_s \mathbb{P}(s_2 = s) \nabla V_{\pi_\theta}(s) \\ &\vdots \end{aligned}$$

Proof of Policy Gradient Theorem: Form 2

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$$\begin{aligned} &= \sum_{(s,a)} \mathbb{P}(s_0 = s, a_0 = a) [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] \\ &+ \gamma \sum_{(s,a)} \mathbb{P}(s_1 = s, a_1 = a) [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] + \gamma^2 \sum_s \mathbb{P}(s_2 = s) \nabla V_{\pi_\theta}(s) \\ &\quad \vdots \\ &= \sum_{t=0}^{\infty} \gamma^t \sum_{(s,a)} \mathbb{P}(s_t = s, a_t = a) [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] \end{aligned}$$

Proof of Policy Gradient Theorem: Form 2

Proof:

$$\begin{aligned} &= \sum_{(s,a)} \mathbb{P}(s_0 = s, a_0 = a) [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] \\ &+ \gamma \sum_{(s,a)} \mathbb{P}(s_1 = s, a_1 = a) [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] + \gamma^2 \sum_s \mathbb{P}(s_2 = s) \nabla V_{\pi_\theta}(s) \\ &\quad \vdots \\ &= \sum_{t=0}^{\infty} \gamma^t \sum_{(s,a)} \mathbb{P}(s_t = s, a_t = a) [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] \\ &= \sum_{(s,a)} [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(x, a)] \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(s_t = s, a_t = a) \end{aligned}$$



Proof of Policy Gradient Theorem: Form 2

Proof:

$$\begin{aligned} &= \sum_{(s,a)} \mathbb{P}(s_0 = s, a_0 = a) [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] \\ &+ \gamma \sum_{(s,a)} \mathbb{P}(s_1 = s, a_1 = a) [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] + \gamma^2 \sum_s \mathbb{P}(s_2 = s) \nabla V_{\pi_\theta}(s) \\ &\quad \vdots \\ &= \sum_{t=0}^{\infty} \gamma^t \sum_{(s,a)} \mathbb{P}(s_t = s, a_t = a) [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] \\ &= \sum_{(s,a)} [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(s_t = s, a_t = a) \\ &= \frac{1}{(1-\gamma)} \sum_{(s,a)} [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] \rho_{\pi_\theta}(s, a) \end{aligned}$$



Proof of Policy Gradient Theorem: Form 2

Proof:

$$\begin{aligned} &= \sum_{(s,a)} \mathbb{P}(s_0 = s, a_0 = a) [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] \\ &+ \gamma \sum_{(s,a)} \mathbb{P}(s_1 = s, a_1 = a) [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] + \gamma^2 \sum_s \mathbb{P}(s_2 = s) \nabla V_{\pi_\theta}(s) \\ &\quad \vdots \\ &= \sum_{t=0}^{\infty} \gamma^t \sum_{(s,a)} \mathbb{P}(s_t = s, a_t = a) [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] \\ &= \sum_{(s,a)} [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(s_t = s, a_t = a) \\ &= \frac{1}{(1-\gamma)} \sum_{(s,a)} [\nabla \log \pi_\theta(s, a) Q_{\pi_\theta}(s, a)] \rho_{\pi_\theta}(s, a) \\ &= \frac{1}{(1-\gamma)} \mathbb{E}_{(s,a) \sim \rho_{\pi_\theta}(\cdot)} [Q_{\pi_\theta}(s, a) \nabla \log \pi_\theta(s, a)] \end{aligned}$$



Estimating Policy Gradient

- We have $\nabla V_{\pi_{\theta}} = (1/(1 - \gamma)) \mathbb{E}_{(s,a) \sim \rho_{\pi_{\theta}}(\cdot, \cdot)} [Q_{\pi_{\theta}}(s, a) \nabla \log \pi_{\theta}(s, a)]$
- How do we estimate $\widehat{\nabla V_{\pi_{\theta}}}$?

Estimating Policy Gradient

- We have $\nabla V_{\pi_{\theta}} = (1/(1 - \gamma)) \mathbb{E}_{(s,a) \sim \rho_{\pi_{\theta}}(\cdot, \cdot)} [Q_{\pi_{\theta}}(s, a) \nabla \log \pi_{\theta}(s, a)]$
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Proposition

$$\begin{aligned} \nabla V_{\pi_\theta} &= (1/(1 - \gamma)) \mathbb{E}_{(s,a) \sim \rho_{\pi_\theta}(\cdot, \cdot)} [Q_{\pi_\theta}(s, a) \nabla \log \pi_\theta(s, a)] \\ &= \mathbb{E}_{\tau \sim P_{\text{traj}, \theta}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t Q_{\pi_\theta}(s_t, a_t) \nabla \log \pi_\theta(s_t, a_t) \right] \end{aligned}$$

Estimating Policy Gradient

Proof:



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Proof:

$$\mathbb{E}_{\tau \sim P_{\text{traj}, \theta}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t Q_{\pi_{\theta}}(s_t, a_t) \nabla \log \pi_{\theta}(s_t, a_t) \right]$$



Estimating Policy Gradient

Proof:

$$\begin{aligned} & \mathbb{E}_{\tau \sim P_{\text{traj}, \theta}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t Q_{\pi_{\theta}}(s_t, a_t) \nabla \log \pi_{\theta}(s_t, a_t) \right] \\ &= \sum_x \sum_a \sum_{t=0}^{\infty} \mathbb{P}(s_t = s, a_t = a) \gamma^t Q_{\pi_{\theta}}(s, a) \nabla \log \pi_{\theta}(s, a) \end{aligned}$$

□

Estimating Policy Gradient

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$$\begin{aligned} & \mathbb{E}_{\tau \sim P_{\text{traj}, \theta}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t Q_{\pi_{\theta}}(s_t, a_t) \nabla \log \pi_{\theta}(s_t, a_t) \right] \\ &= \sum_x \sum_a \sum_{t=0}^{\infty} \mathbb{P}(s_t = s, a_t = a) \gamma^t Q_{\pi_{\theta}}(s, a) \nabla \log \pi_{\theta}(s, a) \\ &= \sum_s \sum_a Q_{\pi_{\theta}}(s, a) \nabla \log \pi_{\theta}(s, a) \left(\sum_{t=0}^{\infty} \mathbb{P}(s_t = s, a_t = a) \gamma^t \right) \end{aligned}$$

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Estimating Policy Gradient

Proof:

$$\begin{aligned} & \mathbb{E}_{\tau \sim P_{\text{traj}, \theta}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t Q_{\pi_{\theta}}(s_t, a_t) \nabla \log \pi_{\theta}(s_t, a_t) \right] \\ &= \sum_x \sum_a \sum_{t=0}^{\infty} \mathbb{P}(s_t = s, a_t = a) \gamma^t Q_{\pi_{\theta}}(s, a) \nabla \log \pi_{\theta}(s, a) \\ &= \sum_s \sum_a Q_{\pi_{\theta}}(s, a) \nabla \log \pi_{\theta}(s, a) \left(\sum_{t=0}^{\infty} \mathbb{P}(s_t = s, a_t = a) \gamma^t \right) \\ &= (1/(1 - \gamma)) \sum_x \sum_a Q_{\pi_{\theta}}(s, a) \nabla \log \pi_{\theta}(s, a) \rho_{\pi_{\theta}}(s, a) \end{aligned}$$

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Estimating Policy Gradient

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□

Estimating Policy Gradient

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□

Estimating Policy Gradient

- We have $\nabla V_{\pi_{\theta}} = (1/(1 - \gamma)) \mathbb{E}_{(x,a) \sim \rho_{\pi_{\theta}}(\cdot, \cdot)} [Q_{\pi_{\theta}}(x, a) \nabla \log \pi_{\theta}(x, a)]$
- How do we estimate $\widehat{\nabla V_{\pi_{\theta}}}$?
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Proposition

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- For a trajectory τ , define:

$$\begin{aligned}\widehat{Q}_{\pi_\theta}(s_t, a_t) &= \sum_{m \geq t}^{\infty} \gamma^{(m-t)} r(s_m, a_m) \\ \widehat{\nabla V_{\pi_\theta}} &= \sum_{t=0}^{\infty} \gamma^t \widehat{Q}_{\pi_\theta}(s_t, a_t) \nabla \log \pi_\theta(s_t, a_t)\end{aligned}$$

Variance Reduction using Baseline

- Advantage function of a policy π is defined as

$$A_{\pi}(s, a) = Q_{\pi}(s, a) - V_{\pi}(s)$$

- Note that $\mathbb{E}_{a \sim \pi(s, \cdot)}[A_{\pi}(s, a)] = 0$

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- Note that $\mathbb{E}_{a \sim \pi(s, \cdot)}[A_{\pi}(s, a)] = 0$

Proposition

$$\nabla V_{\pi_{\theta}} = \frac{1}{(1 - \gamma)} \mathbb{E}_{(s, a) \sim \rho_{\pi_{\theta}}(\cdot, \cdot)} [A_{\pi_{\theta}}(s, a) \nabla \log \pi_{\theta}(s, a)]$$

Variance Reduction using Baseline

Proof:

Let $b_{\pi_\theta}(s)$ be an action independent baseline.

We will first show that $\mathbb{E}_{(s,a) \sim \rho_{\pi_\theta}(\cdot, \cdot)} [b_{\pi_\theta}(s) \nabla \log \pi_\theta(s, a)] = 0$.

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$$\mathbb{E}_{(s,a) \sim \rho_{\pi_\theta}(\cdot, \cdot)} [b_{\pi_\theta}(s) \nabla \log \pi_\theta(s, a)]$$

Variance Reduction using Baseline

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$$\begin{aligned} \mathbb{E}_{(s,a) \sim \rho_{\pi_\theta}(\cdot, \cdot)} [b_{\pi_\theta}(s) \nabla \log \pi_\theta(s, a)] \\ = \mathbb{E}_{s \sim \rho_{\pi_\theta}(\cdot)} [b_{\pi_\theta}(s) \mathbb{E}_{a \sim \pi_\theta(s, \cdot)} [\nabla \log \pi_\theta(s, a)]] \end{aligned}$$

Variance Reduction using Baseline

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Let $b_{\pi_{\theta}}(s)$ be an action independent baseline.

We will first show that $\mathbb{E}_{(s,a) \sim \rho_{\pi_{\theta}}(\cdot, \cdot)} [b_{\pi_{\theta}}(s) \nabla \log \pi_{\theta}(s, a)] = 0$.

$$\begin{aligned} \mathbb{E}_{(s,a) \sim \rho_{\pi_{\theta}}(\cdot, \cdot)} [b_{\pi_{\theta}}(s) \nabla \log \pi_{\theta}(s, a)] \\ &= \mathbb{E}_{s \sim \rho_{\pi_{\theta}}(\cdot)} [b_{\pi_{\theta}}(s) \mathbb{E}_{a \sim \pi_{\theta}(s, \cdot)} [\nabla \log \pi_{\theta}(s, a)]] \\ &= \mathbb{E}_{s \sim \rho_{\pi_{\theta}}(\cdot)} [b_{\pi_{\theta}}(s) \sum_a \pi_{\theta}(s, a) \nabla \log \pi_{\theta}(s, a)] \end{aligned}$$

Variance Reduction using Baseline

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We will first show that $\mathbb{E}_{(s,a) \sim \rho_{\pi_\theta}(\cdot, \cdot)} [b_{\pi_\theta}(s) \nabla \log \pi_\theta(s, a)] = 0$.

$$\begin{aligned} \mathbb{E}_{(s,a) \sim \rho_{\pi_\theta}(\cdot, \cdot)} [b_{\pi_\theta}(s) \nabla \log \pi_\theta(s, a)] &= \mathbb{E}_{s \sim \rho_{\pi_\theta}(\cdot)} [b_{\pi_\theta}(s) \mathbb{E}_{a \sim \pi_\theta(s, \cdot)} [\nabla \log \pi_\theta(s, a)]] \\ &= \mathbb{E}_{s \sim \rho_{\pi_\theta}(\cdot)} [b_{\pi_\theta}(s) \sum_a \pi_\theta(s, a) \nabla \log \pi_\theta(s, a)] \\ &= \mathbb{E}_{s \sim \rho_{\pi_\theta}(\cdot)} [b_{\pi_\theta}(s) \sum_a \nabla \pi_\theta(s, a)] = \mathbb{E}_{s \sim \rho_{\pi_\theta}(\cdot)} [b_{\pi_\theta}(s) \nabla \sum_a \pi_\theta(s, a)] \end{aligned}$$

Variance Reduction using Baseline

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Let $b_{\pi_\theta}(s)$ be an action independent baseline.

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$$\begin{aligned} \mathbb{E}_{(s,a) \sim \rho_{\pi_\theta}(\cdot, \cdot)} [b_{\pi_\theta}(s) \nabla \log \pi_\theta(s, a)] &= \mathbb{E}_{s \sim \rho_{\pi_\theta}(\cdot)} [b_{\pi_\theta}(s) \mathbb{E}_{a \sim \pi_\theta(s, \cdot)} [\nabla \log \pi_\theta(s, a)]] \\ &= \mathbb{E}_{s \sim \rho_{\pi_\theta}(\cdot)} [b_{\pi_\theta}(s) \sum_a \pi_\theta(s, a) \nabla \log \pi_\theta(s, a)] \\ &= \mathbb{E}_{s \sim \rho_{\pi_\theta}(\cdot)} [b_{\pi_\theta}(s) \sum_a \nabla \pi_\theta(s, a)] = \mathbb{E}_{s \sim \rho_{\pi_\theta}(\cdot)} [b_{\pi_\theta}(s) \nabla \sum_a \pi_\theta(s, a)] \\ &= \mathbb{E}_{x \sim \rho_{\pi_\theta}(\cdot, \cdot)} [b_{\pi_\theta}(s) \nabla(1)] = 0 \end{aligned}$$

Variance Reduction using Baseline

Proof:

Let $b_{\pi_\theta}(s)$ be an action independent baseline.

We will first show that $\mathbb{E}_{(s,a) \sim \rho_{\pi_\theta}(\cdot, \cdot)} [b_{\pi_\theta}(s) \nabla \log \pi_\theta(s, a)] = 0$.

$$\begin{aligned} \mathbb{E}_{(s,a) \sim \rho_{\pi_\theta}(\cdot, \cdot)} [b_{\pi_\theta}(s) \nabla \log \pi_\theta(s, a)] &= \mathbb{E}_{s \sim \rho_{\pi_\theta}(\cdot)} [b_{\pi_\theta}(s) \mathbb{E}_{a \sim \pi_\theta(s, \cdot)} [\nabla \log \pi_\theta(s, a)]] \\ &= \mathbb{E}_{s \sim \rho_{\pi_\theta}(\cdot)} [b_{\pi_\theta}(s) \sum_a \pi_\theta(s, a) \nabla \log \pi_\theta(s, a)] \\ &= \mathbb{E}_{s \sim \rho_{\pi_\theta}(\cdot)} [b_{\pi_\theta}(s) \sum_a \nabla \pi_\theta(s, a)] = \mathbb{E}_{s \sim \rho_{\pi_\theta}(\cdot)} [b_{\pi_\theta}(s) \nabla \sum_a \pi_\theta(s, a)] \\ &= \mathbb{E}_{x \sim \rho_{\pi_\theta}(\cdot, \cdot)} [b_{\pi_\theta}(s) \nabla(1)] = 0 \end{aligned}$$

So, for any such baseline $b_\pi(s)$, we get

$$\mathbb{E}_{(s,a) \sim \rho_{\pi_\theta}(\cdot)} [(Q_{\pi_\theta}(s, a) - b_{\pi_\theta}(s)) \nabla \log \pi_\theta(s, a)] = \nabla V_{\pi_\theta}$$

Variance Reduction using Baseline

Proof:

Let $b_{\pi_\theta}(s)$ be an action independent baseline.

We will first show that $\mathbb{E}_{(s,a) \sim \rho_{\pi_\theta}(\cdot, \cdot)} [b_{\pi_\theta}(s) \nabla \log \pi_\theta(s, a)] = 0$.

$$\begin{aligned} \mathbb{E}_{(s,a) \sim \rho_{\pi_\theta}(\cdot, \cdot)} [b_{\pi_\theta}(s) \nabla \log \pi_\theta(s, a)] &= \mathbb{E}_{s \sim \rho_{\pi_\theta}(\cdot)} [b_{\pi_\theta}(s) \mathbb{E}_{a \sim \pi_\theta(s, \cdot)} [\nabla \log \pi_\theta(s, a)]] \\ &= \mathbb{E}_{s \sim \rho_{\pi_\theta}(\cdot)} [b_{\pi_\theta}(s) \sum_a \pi_\theta(s, a) \nabla \log \pi_\theta(s, a)] \\ &= \mathbb{E}_{s \sim \rho_{\pi_\theta}(\cdot)} [b_{\pi_\theta}(s) \sum_a \nabla \pi_\theta(s, a)] = \mathbb{E}_{s \sim \rho_{\pi_\theta}(\cdot)} [b_{\pi_\theta}(s) \nabla \sum_a \pi_\theta(s, a)] \\ &= \mathbb{E}_{x \sim \rho_{\pi_\theta}(\cdot, \cdot)} [b_{\pi_\theta}(s) \nabla(1)] = 0 \end{aligned}$$

So, for any such baseline $b_\pi(s)$, we get

$$\mathbb{E}_{(s,a) \sim \rho_{\pi_\theta}(\cdot)} [(Q_{\pi_\theta}(s, a) - b_{\pi_\theta}(s)) \nabla \log \pi_\theta(s, a)] = \nabla V_{\pi_\theta}$$

This is in particular true for the baseline $b_{\pi_\theta}(x) = V_{\pi_\theta}(x)$

Policy Gradient Theorem(s)

Theorem

The following as expressions for ∇V_{π_θ} :

$$\nabla V_{\pi_\theta} = \mathbb{E}_{\tau \sim P_{\text{traj}, \theta}(\cdot)} \left[R(\tau) \sum_{t=0}^{\infty} \nabla \log \pi_\theta(s_t, a_t) \right] \quad (1)$$

$$\nabla V_{\pi_\theta} = \frac{1}{(1 - \gamma)} \mathbb{E}_{(s, a) \sim \rho_{\pi_\theta}(\cdot, \cdot)} [Q_{\pi_\theta}(s, a) \nabla \log \pi_\theta(s, a)] \quad (2)$$

$$\nabla V_{\pi_\theta} = \mathbb{E}_{\tau \sim P_{\text{traj}, \theta}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t Q_{\pi_\theta}(s_t, a_t) \nabla \log \pi_\theta(s_t, a_t) \right] \quad (3)$$

$$\nabla V_{\pi_\theta} = \frac{1}{(1 - \gamma)} \mathbb{E}_{(s, a) \sim \rho_{\pi_\theta}(\cdot, \cdot)} [A_{\pi_\theta}(s, a) \nabla \log \pi_\theta(s, a)] \quad (4)$$

Actor-Critic Algorithm

- We have, $\nabla V_{\pi_{\theta}} = \mathbb{E}_{(s,a) \sim \rho_{\pi_{\theta}}(\cdot, \cdot)} [Q_{\pi_{\theta}}(s, a) \nabla \log \pi_{\theta}(s, a)]$

Actor-Critic Algorithm

- We have, $\nabla V_{\pi_{\theta}} = \mathbb{E}_{(s,a) \sim \rho_{\pi_{\theta}}(\cdot, \cdot)} [Q_{\pi_{\theta}}(s, a) \nabla \log \pi_{\theta}(s, a)]$
- How do we get $Q_{\pi_{\theta}}(s, a)$?

Actor-Critic Algorithm

- We have, $\nabla V_{\pi_\theta} = \mathbb{E}_{(s,a) \sim \rho_{\pi_\theta}(\cdot, \cdot)} [Q_{\pi_\theta}(s, a) \nabla \log \pi_\theta(s, a)]$
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 - ▶ θ is changing. So, w should also change with respect to θ

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- **Critic:** Updates Q-value parameter w
 - ▶ (*Evaluate* the policy corresponding to θ)
- **Actor:** Updates policy parameter θ
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for each step **do**

$$\theta = \theta + \alpha_{\theta} \nabla_{\theta} \log(\pi_{\theta}(s, a))$$

Sample the next state s' , sample the next action $a' \sim \pi_{\theta}(s', \cdot)$

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- This is an example of **two timescale stochastic approximation algorithm**